Mayank Bharwal

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Education

University of Toronto - St. George **Expected Graduation: May 2027**
Bachelor of Computer Science — AI and Systems Double Focus *Toronto, ON*

- Dean's List Scholar x2
- Courses: Computer Architecture, Systems Programming, Machine Learning, Web Programming, Software Consulting

Work Experience

TD Bank **May 2026 – Present**
AI Software Engineering Intern *Toronto, ON*

- Awarded Innovation Trophy during Innovation Week within two weeks of joining for developing the competition's most innovative solution
- Contributed to a 7-engineer team building a patent-pending autonomous AI platform for enterprise application monitoring, testing, incident response, and operational resilience across multiple business units
- Engineered agentic workflows using LangGraph, LangChain, FastAPI, Selenium, Playwright, and Python to continuously validate hundreds of applications, with projected 5× reduction in manual testing effort and 10× improvement in MTTR
- Built scalable services and observability infrastructure using Spring Boot, Docker, Kubernetes, Linux, Prometheus, and Grafana, supporting a platform projected to deliver $25M+ in annual operational savings

University of Toronto's Amateur Rocketry Club **Aug 2025 – Present**
Software Engineering Lead *Toronto, ON*

- Architected SAL, HAL, and TEST layers using Cython, design patterns, and JSBSim, achieving 95%+ simulation fidelity and 60% improvement in landing precision through real-time sensor fusion
- Engineered telemetry stack with radio communication and ground station GUI, enabling live monitoring and data logging for flight systems with 95% accuracy
- Managed 8–10 developers using Agile workflow, implementing GitHub Actions CI/CD with hardware-in-the-loop testing to streamline development and deployment

Trustworthy Machine Intelligence **Oct 2025 – Present**
AI Engineer *Toronto, ON*

- Simulated 50+ LLM agents in small-world networks using Python, achieving 35% higher cooperation in Public Goods games over 100 rounds
- Designed agent architecture with memory, personality, and LLM-based decision policies, improving strategic coordination accuracy by 28% across repeated Prisoner's Dilemma interactions
- Applied interventions (penalties, subsidies, cross-community links), increasing fairness metrics by 22% and reducing polarization by 15%

Hyundai Motors **Jun 2023 – Aug 2023**
Full Stack Software Engineering Intern *India*

- Designed and implemented front end and backend of employee management software using HTML, CSS, JavaScript and Django, deployed on company's internal server
- Created SQLite database for storing employee information including performance history and projects history
- Developed front end using React.js to make information flow more interactive and user friendly
- Collected data from 200+ employees, achieving significant reduction in data collection time through automated forms

Leadership

Computer Science Tutor | *UrbanPro.com* **Mar 2020 – Oct 2020**

- Tutored 10 students in Java and CS concepts; 9/10 scored above 95/100 in board exams

Technical Skills

Languages: Python, Java, C++, C, JavaScript, TypeScript, SQL, Rust, HTML/CSS
Frameworks/Libraries: React, Node.js, PyTorch, TensorFlow, scikit-learn, Django, FastAPI, MongoDB, PostgreSQL
Tools/Platforms: Docker, AWS, Git, Kubernetes, Jenkins, Google Cloud, Linux, Bash, REST APIs, GraphQL